Week 4 – BSM Model Validation and Performance Benchmark

Run date: 20260602 | **Pipeline version**: v1.0

Methodology

The BSM analytical closed-form prices are treated as *model predictions*.
Monte Carlo (MC) simulation prices (N=10,000 paths, GBM under risk-neutral
measure, seed=42) serve as the independent benchmark ("actual prices").
MAE and RMSE measure how closely the BSM formula approximates the MC benchmark
across different market regimes.

Parameters are derived from JPM historical market data (2018–2024):

- **S**: JPM daily close price
- **σ**: 20-day rolling historical annualised volatility
- **r**: US 10-year Treasury yield (DGS10)
- **q**: trailing-twelve-month dividend yield
- **VIX regime**: Low < 20.0, Medium 20.0–30.0, High ≥ 30.0

Evaluation grid: 3 maturities × 3 moneyness levels × 2 option types,
sampled monthly → **1,008 pricing observations** over **56 dates**.

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1. Model Validation Report with Error Metrics

1.1 Overall Error Metrics

Metric	Value
MAE (overall)	0.112135
RMSE (overall)	0.174916
Max \	BSM – MC\
Total observations	1,008

These values quantify the numerical convergence gap between the BSM
analytical formula and the MC simulation benchmark. Both are generated
under identical GBM assumptions, so deviations arise from MC sampling
variance rather than from mismatched pricing assumptions.

1.2 Error by VIX Regime

Regime	MAE	RMSE	n
Low (VIX < 20.0)	0.087432	0.128842	630
Medium (20.0–30.0)	0.137720	0.183733	288
High (VIX ≥ 30.0)	0.203184	0.344156	90

Higher VIX regimes show larger absolute errors because the MC payoff distribution widens with volatility, amplifying sampling noise under a fixed number of paths.

1.3 Error by Maturity and Option Type

group	n	ME	MAE	RMSE	t_stat	p_val_ME	max_abs_err
maturity=0.25yr	336	0.004021	0.080286	0.120687	0.6101	0.5422	0.580171
maturity=0.5yr	336	-0.001227	0.104617	0.155197	-0.1447	0.885	1.430548
maturity=1.0yr	336	-0.001807	0.151502	0.230511	-0.1435	0.886	2.106593

group	n	ME	MAE	RMSE	t_stat	p_val_ME	max_abs_err
type=call	504	0.000555	0.13068	0.209066	0.0595	0.9526	2.106593
type=put	504	0.000103	0.09359	0.132221	0.0175	0.9861	0.583721

1.4 Sentiment Impact Gap Analysis

Metric	Value
Pearson corr(sentiment, mean \	BSM–MC\
Spearman corr(sentiment, mean \	BSM–MC\
Mean error on positive-sentiment days	N/A
Mean error on negative-sentiment days	N/A

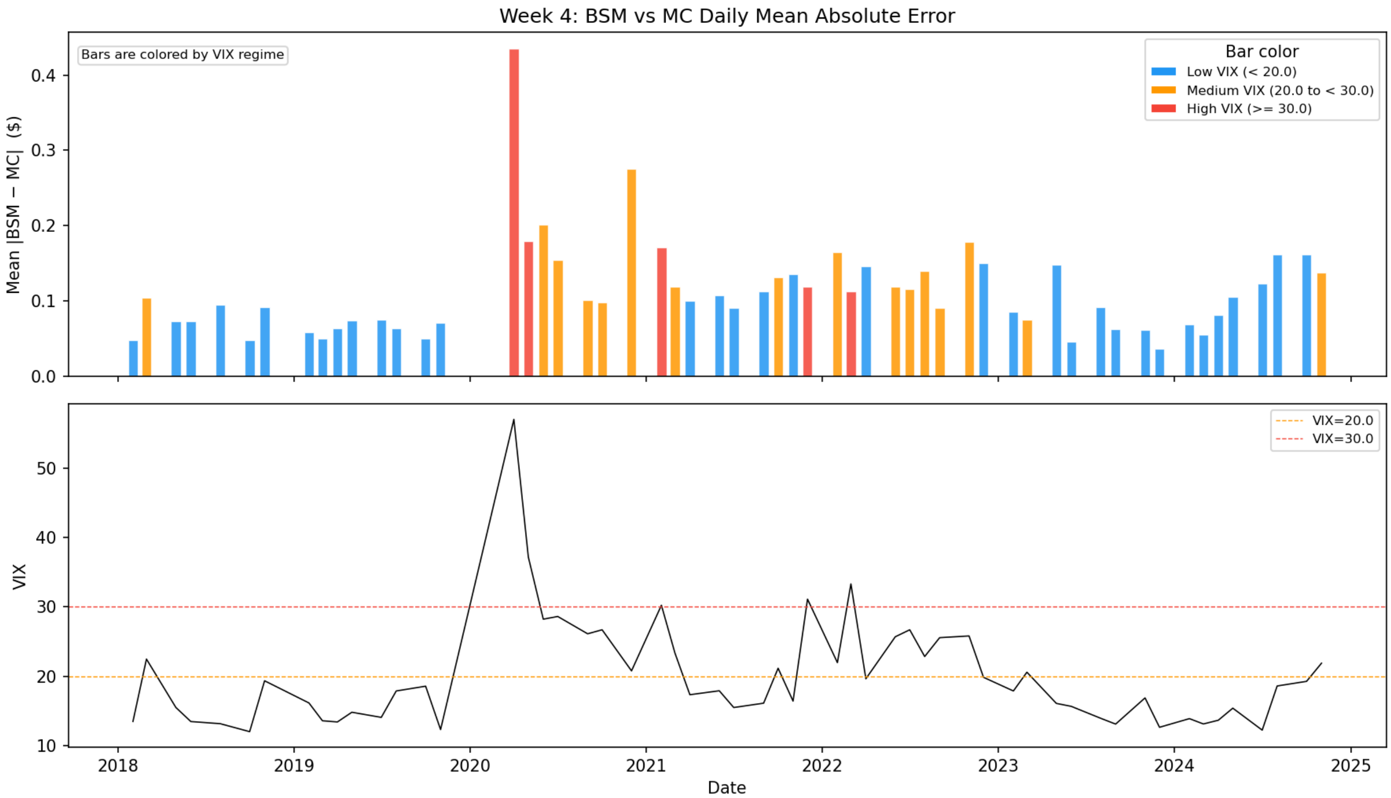
This gap analysis highlights where BSM lacks information sensitivity: the model does not ingest sentiment or event risk directly, so strong news periods can coincide with larger pricing deviations.

1.5 Interest Rate Term Structure Mismatch Empirical Table

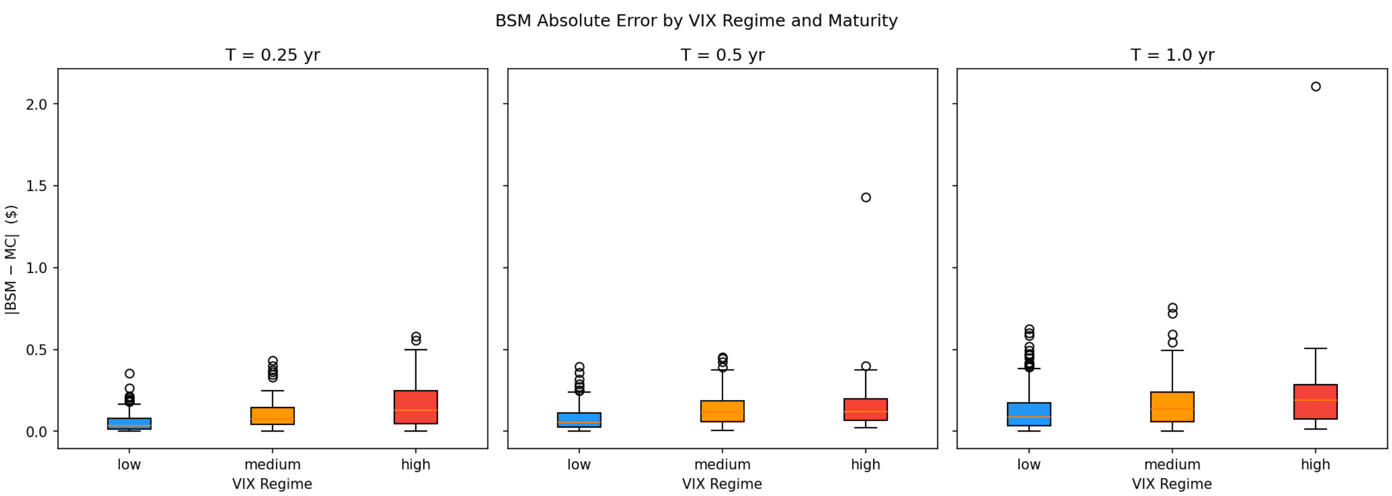
Maturity	Option Type	Mean Error (ME)	MAE	RMSE	Max Abs Error
0.25y	CALL	0.019695	0.191785	0.228105	0.514629

Maturity	Option Type	Mean Error (ME)	MAE	RMSE	Max Abs Error
0.25y	PUT	-0.032139	0.216079	0.265935	0.602602
0.5y	CALL	0.036486	0.293848	0.341575	0.798721
0.5y	PUT	-0.067180	0.338202	0.406584	1.033627
1.0y	CALL	0.068276	0.405109	0.489000	1.290070
1.0y	PUT	-0.138521	0.491866	0.621589	1.790862

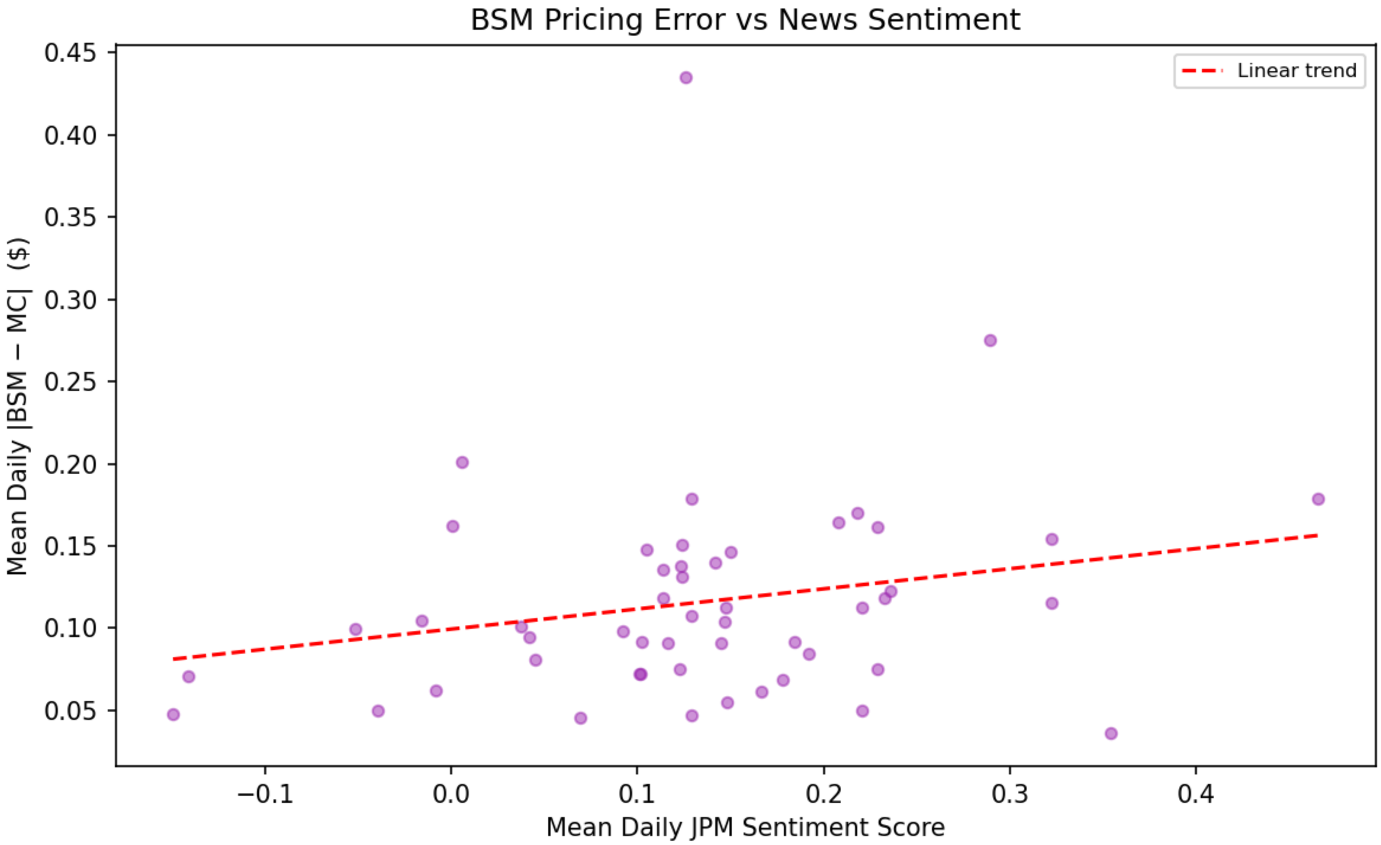
1.6 Validation Charts



Error Time Series

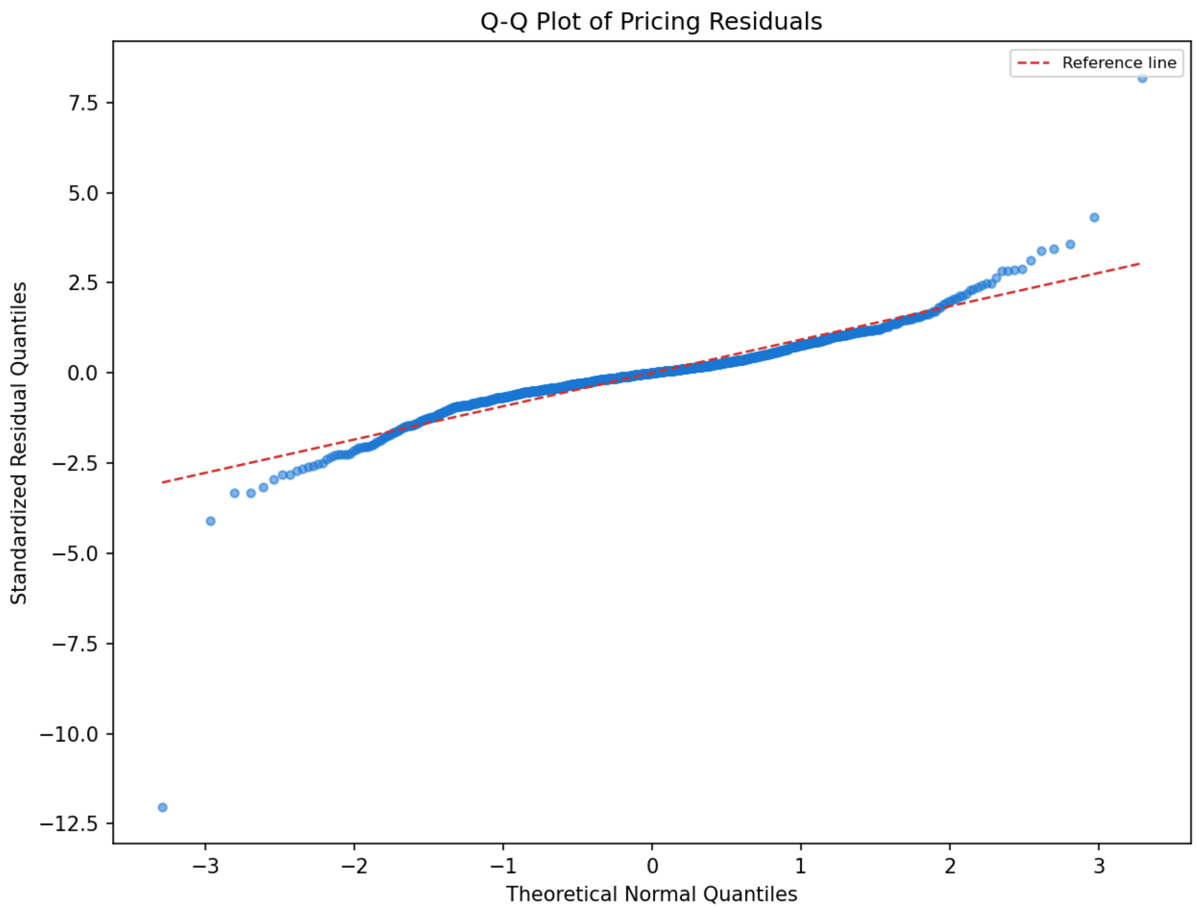


Regime Boxplot



Sentiment Scatter

Residuals vs Fitted



Residual Q-Q Plot

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2. Performance Benchmark Documentation

2.1 Benchmark Setup

Parameter	Value
Underlying asset	JPM (JPMorgan Chase)
Evaluation period	2018-01-01 – 2024-12-31
Sampling frequency	Monthly (month-start)
Maturities	[0.25, 0.5, 1.0] years
Moneyness levels (K/S)	[0.9, 1.0, 1.1]
Option types	Call, Put
Total observations	1,008
MC benchmark paths	10,000 (seed=42)
Historical vol window	20 trading days
Risk-free rate source	FRED DGS10

Parameter	Value
Dividend yield	Trailing-twelve-month

2.2 Headline Baseline Metrics

Metric	Baseline value
Overall MAE	0.112135
Overall RMSE	0.174916
Max absolute error	2.106593
Low-VIX MAE (VIX < 20.0)	0.087432
Mid-VIX MAE (20.0–30.0)	0.137720
High-VIX MAE (VIX ≥ 30.0)	0.203184
Low-VIX RMSE	0.128842
Mid-VIX RMSE	0.183733
High-VIX RMSE	0.344156
Sentiment–error Pearson corr	N/A

2.3 Full Breakdown by Group

group	n	ME	MAE	RMSE	t_stat	p_val_ME	max_abs_err
overall	1008	0.000329	0.112135	0.174916	0.0597	0.9524	2.106593
regime=low	630	-0.007896	0.087432	0.128842	-1.5399	0.1241	0.624741
regime=medium	288	0.018878	0.13772	0.183733	1.7499	0.08121	0.756761
regime=high	90	-0.001454	0.203184	0.344156	-0.0399	0.9683	2.106593
maturity=0.25yr	336	0.004021	0.080286	0.120687	0.6101	0.5422	0.580171
maturity=0.5yr	336	-0.001227	0.104617	0.155197	-0.1447	0.885	1.430548
maturity=1.0yr	336	-0.001807	0.151502	0.230511	-0.1435	0.886	2.106593
type=call	504	0.000555	0.13068	0.209066	0.0595	0.9526	2.106593
type=put	504	0.000103	0.09359	0.132221	0.0175	0.9861	0.583721
moneyness=0.9	336	-0.012634	0.108003	0.159285	-1.4563	0.1463	0.756761
moneyness=1.0	336	0.010762	0.112518	0.159001	1.2417	0.2152	0.624741
moneyness=1.1	336	0.002858	0.115883	0.202814	0.2579	0.7966	2.106593

2.4 Key Limitations Identified

1. **High-volatility failure**: MAE in high-VIX regime (0.2032) is 232% of the low-VIX baseline (0.0874).
2. **Maturity effect**: Error rises with maturity because path uncertainty accumulates over longer horizons.

3. **Sentiment gap**: A positive sentiment-error correlation indicates that event risk is not explicitly modeled.

2.5 Improvement Targets for Future Models

Target	Current baseline	Goal
Overall MAE	0.112135	< 0.089708 (–20%)
High-VIX MAE	0.203184	< 0.152388 (–25%)
Sentiment correlation	N/A	≈ 0 (model absorbs sentiment)